

Participation Gradients Discipline Coordination Traps: Agency, Social Cohesion, and Activation Policy in a Heterogeneous-Agent Economy

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Abstract

I model participation in social life as a discrete choice with social interactions inside an incomplete-markets economy, in which agency, the capacity to convert effort into income, differs across education groups. The main result is a bound that turns ordinary participation statistics into a test for coordination traps. With lognormal belonging tastes, the slope of the aggregate participation map depends on the taste distribution only through its dispersion and the participation rates that distribution induces, so multiple equilibria require taste dispersion below a threshold computable from observed group rates and one preference share, with no structural estimation. The bound tightens as the participation gradient steepens: an observed education gradient is evidence against coordination traps rather than neutral with respect to them. At the French moments the bound is 0.45 against the dispersion of 0.50 needed to fit the gradient, so the calibrated economy sits about ten percent outside the coordination region, and the model's own numerical boundary near 0.41 confirms the ordering. Three results follow. The education gradient in participation divides evenly between the taste for belonging and agency operating through wealth, the income effect beating the price of time, which is why participation rises with income although richer households face a higher opportunity cost of time. A budget-balanced make-work-pay subsidy has a direct effect of eight percentage points and an equilibrium effect of twenty-nine, from 37 percent participation to 8, so outside the multiplicity region the complementarity still amplifies policy by a factor of 3.6. And a participation tax credit, the analogue of France's charitable-donations deduction and the United Kingdom's Gift Aid, moves participation the other way at lower fiscal cost, so two policies from the same pro-work family have opposite social consequences and appear as mirror images on a GDP and a GDP-B ledger that single-index evaluation cannot separate.

1 Introduction

The companion paper to this one embeds social cohesion in a heterogeneous-agent economy and establishes two things: that a recognisable activation policy buys consumption at the cost of the

*Draft. Companion to "Wellbeing and the Social Fabric in a Heterogeneous-Agent Economy"; the two papers share one Julia engine, and every number and figure here is reproduced by the scripts `sa_main.jl` and `sa_figures.jl` in the public repository.

social fabric, with a regressive incidence invisible to single-index evaluation, and that the smooth social-multiplier route to multiple equilibria does not survive measured labour-supply elasticities. The second finding is constructive: it says that a credible account of social coordination needs a margin whose aggregate response can be steep without an implausible intensive-margin elasticity, and the theory of social interactions has long identified that margin as a discrete choice with complementarities [Brock and Durlauf, 2001]. This paper builds it, and puts agency, the second SAGE dimension [Lima de Miranda and Snower, 2020], at its centre.

The discrete choice is participation in social life. One joins the association, the club, the standing commitment, or one does not; joining means committing a minimum meaningful share of time. This is not a modelling flourish but what the data measure: in the French statistical office’s long-running series, 42 percent of residents belong to at least one association, the rate rises steeply with education, 56 percent above the baccalaureat against 22 percent with no diploma, and the pattern has been stable for thirty years [ins, 2016]. Time-use data put participants’ associative and social commitment at around a tenth of committed time [Roy, 2012]. The model is calibrated to exactly these moments: an aggregate participation rate of a third and group rates of one quarter and forty-five hundredths for the lower- and higher-education groups.

Four results follow.

First, and this is the paper’s main contribution, the coordination question is answered with a theorem rather than a scan. In the threshold version of the model with lognormal belonging tastes, the slope of the aggregate participation map has a closed form in which the taste distribution enters only through its dispersion σ_m and the participation rates that distribution induces. Multiple equilibria then require σ_m to fall below a bound assembled from observed group participation rates and the private share of the participation payoff, and from nothing else: no structural estimation, no unobserved parameter. The bound has a property that gives it teeth. It is largest when groups participate at one half and shrinks as rates move apart, so the steeper the observed participation gradient, the harder coordination traps are to sustain. A visible gradient is not neutral evidence about traps; it is evidence against them. At the French moments the bound is 0.45 while fitting the gradient requires 0.50, so the economy sits about ten percent outside the coordination region, and because the threshold version overstates the true slope the conclusion is conservative. Scanning the full model puts its actual boundary near 0.41, as the proposition’s $1/\sigma_m$ scaling predicts. This is the honest quantitative answer to whether left-behind communities are coordination failures: in this model, disciplined by participation data, they are not, though the economy lives near the edge where they could be. It is also portable, because any researcher holding a participation gradient can compute the bound for their own setting.

Second, the model reproduces the education gradient in participation and says where it comes from. Half the gap is the taste for belonging, which the underlying wellbeing indicators put higher among the more educated. The other half is agency, and it works through wealth: higher agency makes households richer, richer households have a lower marginal utility of consumption, and so the fixed time commitment costs them less in utility terms. The income effect beats the price-of-time

effect. This resolves, inside a structural model, the apparent puzzle of why participation rises with income although richer people face a higher opportunity cost of time, the same force that makes volunteering an increasing function of education in every survey. It also supplies the input the first result needs: the gradient is real, and it is evidence of dispersed tastes.

Third, the activation-policy result of the companion paper becomes dramatic on the participation margin. A budget-balanced 20 percent make-work-pay subsidy raises the return to market hours and taxes everyone a lump sum to pay for it; both push against the time commitment. The direct effect, holding everyone else's participation fixed, is a fall of eight percentage points. The equilibrium effect is a fall of twenty-nine, from 37 percent participation to 8: the social feedback multiplies the direct effect three and a half times, because each household's exit lowers the belonging return of every other household. The same complementarity that could not fold the smooth model amplifies policy shocks on the discrete margin, and the amplification is the quantitatively dominant part of the social cost.

Fourth, the contrast that makes the policy point usable. Raising the lower-education group's agency to the population mean, an empowerment policy in the SAGE sense, raises equilibrium participation from 37 to 51 percent, with both groups participating more. A participation tax credit, modelled at the real rebate rates of the United Kingdom's Gift Aid and France's charitable-donations deduction, does the same from the other direction and more cheaply: Gift Aid's 25 percent rebate lifts participation to 83 percent at a fifth of the work subsidy's fiscal cost. Because such credits are claimed unevenly, I add a take-up gradient, and the result runs against the natural worry. The credit remains equalising at every take-up rate I try: the ratio of higher- to lower-group participation falls from 1.79 at baseline to between 1.01 and 1.51, so imperfect take-up makes the instrument less equalising rather than regressive. Policies from the same pro-work family therefore have opposite effects on the social fabric: subsidising hours raises the price of time and unravels participation, while empowering people or paying them for the act of joining crowds it in. An evaluation framework that does not carry the social dimension cannot tell these policies apart on the margin where they differ most.

The paper proceeds as follows. Section 2 lays out the model and its relation to the shared engine. Section 3 calibrates. Section 4 presents the gradient and its decomposition, which establishes the dispersion the main result needs. Section 5 states and proves the heterogeneity-multiplicity trade-off and applies it to the French data. Section 6 runs the three policies and sets them on the two ledgers. Section 7 concludes with the welfare reading and the road to general equilibrium.

The headline calibration is for France because the supporting French data is unusually complete. The model engine carries seven country rows (France, Germany, Italy, the United States, Colombia, South Africa, and China) populated from the same harmonised cross-country sources: OECD How's Life indicators by education for the agency and belonging gradients, the World Values Survey Wave 7 for the participation gradient, the Multinational Time Use Study for the time-use share, and the Open Inequality Atlas [Conway, 2025] for the wealth-Gini target. The atlas reconciles the World Inequality Database, the ECB Household Finance and Consumption Survey, the Luxembourg

Wealth Study, the Survey of Consumer Finances, and the Federal Reserve Distributional Financial Accounts for 213 countries from 1800 to 2025, with comparability tiers per observation, which is what makes a single calibration pipeline genuinely portable across countries that vary widely in inequality. Robustness statements throughout the paper use the country rows; the cross-country fit table is in the public repository.

2 Model

2.1 Households

The economy consists of two permanent education groups $g \in \{l, h\}$ of equal size, the clean separation of education from income risk: agency α_g and the belonging taste B_g are constant within a group, while idiosyncratic productivity z follows the same two-state Markov process within each group as in the companion paper. Within each group, households differ permanently in a belonging taste m , drawn from a lognormal distribution with unit mean and log standard deviation σ_m ; tastes for social life are persistent traits, in the spirit of the persistent cooperator types measured by Fischbacher et al. [2001].

Each period a household chooses consumption, savings $a' \geq 0$ at gross rate R , market effort e , and a participation indicator $d \in \{0, 1\}$. Participating commits a minimum meaningful time lump $\bar{q} = 0.10$ of the time endowment, so feasible effort satisfies $e + \bar{q}d \leq 1$, and total committed time $T = e + \bar{q}d$ carries the usual convex disutility. The budget is

$$c + a' = R a + (1 + \tau) \alpha_g e z - T_x, \quad a' \geq 0, \quad (1)$$

with τ the make-work-pay subsidy and T_x the lump-sum tax that finances it.

2.2 The participation payoff and the fabric

The social fabric is participation time, $Q = \bar{q} \cdot P(d = 1)$, so the aggregate state is summarised by the participation rate. This replaces the companion paper's all-non-work-time fabric with the cleaner object the time-use and participation data actually measure; a free rider's leisure no longer builds fabric. The payoff to participating, per unit of committed lump, is

$$\kappa \Lambda B_g m (\omega + (1 - \omega) P(d = 1)), \quad (2)$$

where κ is the interaction strength, Λ the cohesion weight carried over from the shared engine, and $\omega \in [0, 1]$ the private share of the payoff: the warm-glow of participating that does not depend on how many others participate. The private share anchors participation away from zero, as it must, since people volunteer even in low-participation places; the complementarity share $(1 - \omega)$ is the Brock and Durlauf [2001] interaction that makes participation feed on itself. The decision-utility versus experienced-wellbeing architecture of the companion paper is unchanged: the payoff above

moves behaviour, while the wellbeing dashboard prices the enjoyed fabric.

2.3 Equilibrium and computation

A stationary equilibrium is a participation rate equal to the participation it induces. Define the aggregate map $G(r)$: given a conjectured rate r , each (group, taste) household solves its dynamic programme, and $G(r)$ is the implied population participation rate; equilibria are crossings $G(r) = r$, stable where the map crosses from above. Computation exploits a reduction: within a group, the payoff depends on $(\kappa, m, B_g, \omega, r)$ only through one scalar, so each group’s response is precomputed once on a grid of that scalar at production resolution, and every map evaluation, calibration loop, and policy counterfactual is interpolation. This removes at source the near-fold grid sensitivity that a direct fixed-point probe suffers, and it makes the full parameter-space scan of Section 5 feasible. The household problem itself is the discrete dynamic programme of the shared engine, with the participation indicator added to the choice set and the stationary distribution computed exactly.

3 Calibration

Everything inherited from the companion paper stays at its sourced value: preferences (risk aversion 2, Frisch one half, the effort scale tied to the INSEE time-use work share), the income process, the discount factor and interest rate, agency $\alpha_l = 0.765$, $\alpha_h = 0.911$ and tastes $B_l = 0.80$, $B_h = 0.94$ from the OECD Better Life indicators by education, and the cohesion weight Λ . The participation lump $\bar{q} = 0.10$ is the time-use order of magnitude for participants’ associative commitment. That leaves three new objects: the interaction strength κ , the taste dispersion σ_m , and the private share ω .

I calibrate (κ, σ_m) to the two INSEE participation moments, group rates of 0.25 and 0.45 at a stable equilibrium, with $\omega = 0.30$ held at its central value and swept in robustness. The fit is tight: at $\kappa = 10.0$, $\sigma_m = 0.50$, the economy has a stable equilibrium with aggregate participation 0.371 and group rates 0.265 and 0.476, against targets of 0.25 and 0.45 (Figure 1). The private share ω has no clean point estimate; it is swept over $\{0.15, 0.30, 0.50\}$ and results are reported with that sensitivity stated. All numbers below are robust to doubling the number of taste nodes, the check that the cruder prototype of the companion paper failed.

4 The gradient: taste or agency?

Why does participation rise with education? The model offers two channels and the calibration lets us weigh them. The taste channel is direct: $B_h > B_l$ in the wellbeing indicators, so the more educated value belonging more. The agency channel is indirect and runs through wealth: higher α converts effort into income at a better rate, higher income builds wealth, wealth lowers the marginal utility of consumption, and a cheaper marginal util makes the fixed time lump less costly to bear. Note the sign battle inside the agency channel: higher agency also raises the price of an hour, which

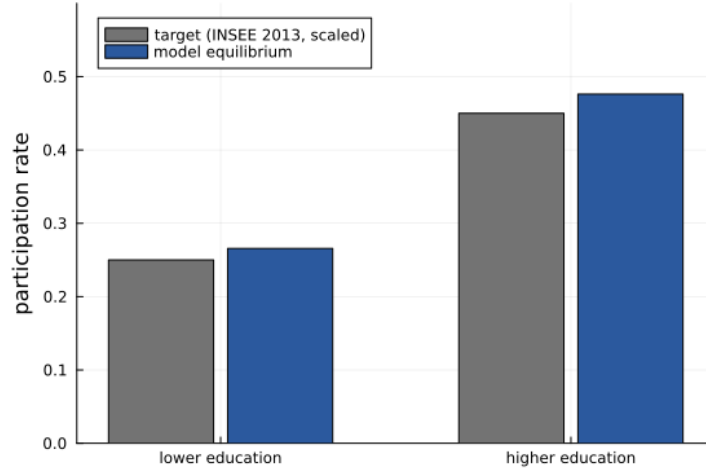


Figure 1: Participation by education group: the INSEE-based calibration targets and the model’s calibrated equilibrium.

pushes participation down. At the consensus labour-supply elasticity the income effect wins, which is what the data demand, since participation in fact rises with income.

Shutting each channel down in turn at the calibrated point decomposes the gap of 21 percentage points: with tastes equalised the agency channel alone produces a gap of 10 points; with agency equalised the taste channel alone produces 10 points (Figure 2). The gradient is, in this model, half who people are and half what they can afford, and the affordability half is a wealth effect, not a price effect. The distributional reading matters: the lower-education group participates less not only because it values belonging less on the survey measures, but because participation is a normal good and the group is poorer.

5 Heterogeneity against multiplicity

The participation margin restores what the smooth model could not sustain: for concentrated taste distributions the aggregate map folds and the economy has genuinely distinct stable equilibria. Whether the French economy sits in that region is a quantitative question, and it turns out to have an analytical answer. The answer is that the observed participation gradient is itself the evidence that rules multiplicity out.

5.1 The slope of the aggregate map

Abstract for a moment from within-group wealth heterogeneity, so that households in a group differ only in their belonging taste m . A household participates whenever its payoff exceeds the utility cost of the time lump, a cost C_g common within the group, and this defines a taste threshold

$$m_g^*(r) = \frac{C_g}{\kappa \Lambda B_g (\omega + (1 - \omega) r)}.$$

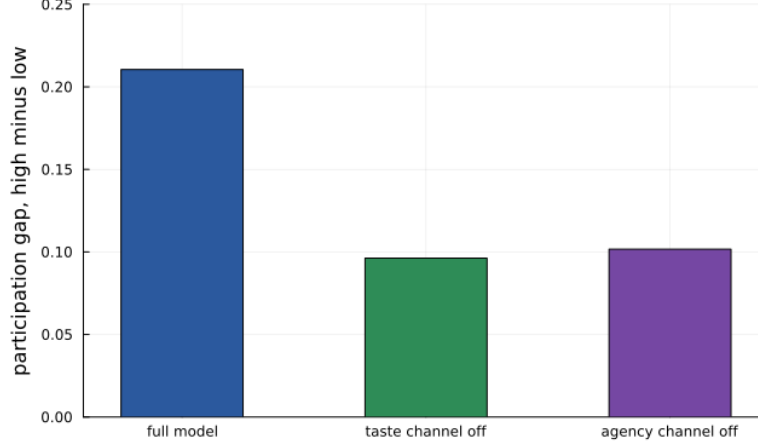


Figure 2: The education gap in participation at the calibrated equilibrium, and the gap with each channel shut down. Taste and agency-through-wealth each carry about half.

The threshold falls as participation rises, which is the complementarity. Group participation is $p_g(r) = 1 - F(m_g^*(r))$ with F the taste distribution, and the aggregate map is $G(r) = \sum_g s_g p_g(r)$. Differentiating, and using $dm_g^*/dr = -m_g^*(1 - \omega)/(\omega + (1 - \omega)r)$,

$$G'(r) = \frac{1 - \omega}{\omega + (1 - \omega)r} \sum_g s_g m_g^*(r) f(m_g^*(r)).$$

Everything about the taste distribution reaches the aggregate slope through the single object $mf(m)$, and for the lognormal that object takes a form which depends only on the participation rate the distribution implies.

Proposition 1 (Heterogeneity-multiplicity trade-off). *Let tastes be lognormal with log standard deviation σ_m . At any equilibrium with aggregate rate r and group rates $\{p_g\}$,*

$$G'(r) = \frac{1 - \omega}{\omega + (1 - \omega)r} \cdot \frac{1}{\sigma_m} \cdot \sum_g s_g \varphi(\Phi^{-1}(1 - p_g)),$$

with φ and Φ the standard normal density and distribution function. Multiple equilibria therefore require

$$\sigma_m < \bar{\sigma} \equiv \frac{1 - \omega}{\omega + (1 - \omega)r} \sum_g s_g \varphi(\Phi^{-1}(1 - p_g)),$$

a bound determined entirely by observed participation rates and the private share. Moreover $\bar{\sigma} \leq [(1 - \omega)/(\omega + (1 - \omega)r)]\varphi(0)$, with equality only if every group participates at one half, so the bound tightens as group rates move away from one half and away from each other.

Proof. For $\ln m \sim N(\mu, \sigma_m^2)$ the density satisfies $f(m) = \varphi((\ln m - \mu)/\sigma_m)/(m\sigma_m)$, so $mf(m) = \varphi(z)/\sigma_m$ with $z = (\ln m - \mu)/\sigma_m$. At the threshold, $p_g = 1 - \Phi(z_g)$ gives $z_g = \Phi^{-1}(1 - p_g)$, hence $m_g^*f(m_g^*) = \varphi(\Phi^{-1}(1 - p_g))/\sigma_m$, and substitution gives the first display; note that the location μ

drops out. Multiple equilibria require the aggregate map to cross the forty-five degree line from below at some point, hence $G' > 1$ there, which rearranges to the bound. The last claim follows from $\varphi(\Phi^{-1}(1-p)) \leq \varphi(0)$, with equality if and only if $p = 1/2$. $\square$

The proposition says something that is not evident before the algebra: the aggregate slope depends on the taste distribution only through its dispersion and through the participation rates that distribution induces. Two economies with the same observed rates and the same private share have the same aggregate slope, whatever else differs between them. Participation rates, which statistical offices report routinely, are therefore sufficient to bound the region in which coordination traps can live.

The trade-off in the name is the tension between the two jobs the dispersion has to do. Multiplicity needs the marginal household to be one of many, so it needs tastes concentrated, σ_m small. An observed education gradient, with groups participating at materially different and interior rates, needs tastes spread, σ_m large. The final inequality sharpens the point: the steeper the gradient, the further the group rates lie from one half, the smaller the density term, and the tighter the bound. A visible participation gradient is not neutral evidence about coordination traps. It is evidence against them.

5.2 What the bound says about France

At the calibrated equilibrium the aggregate rate is $r = 0.371$, group rates are 0.265 and 0.476, and $\omega = 0.30$. The complementarity factor is 1.251 and the density term 0.363, so $\bar{\sigma} = 0.454$. The dispersion required to fit the observed gradient is $\sigma_m = 0.50$, which exceeds the bound. Multiplicity is ruled out, and by an economically small margin: the calibrated economy sits about ten percent outside the boundary measured in taste dispersion.

Because the proposition abstracts from within-group wealth heterogeneity, which smooths each group's response and so lowers the true slope, it delivers a necessary condition rather than a sufficient one. The model's own numbers confirm the direction. At the calibrated point the map's actual slope is $G'(r^*) = 0.81$ against the proposition's 0.91, both below one. Scanning the full model directly, multiple stable equilibria appear for $\sigma_m \leq 0.40$ and are gone by $\sigma_m = 0.42$, placing the true boundary near 0.41, inside the analytical bound as it must be; and the $1/\sigma_m$ scaling of the proposition predicts that boundary to two digits, since $0.5 \times 0.81 = 0.40$. Figure 3 shows the same conclusion in the two-dimensional scan of the (κ, σ_m) plane, where the multiplicity region sits at low dispersion, the data-fitting region at higher dispersion, and the two are adjacent and disjoint. The same scan at $\omega = 0.15$ and $\omega = 0.50$ gives the same answer: at low ω multiplicity is widespread but nothing in it fits the moments, and at high ω multiplicity disappears entirely.

Three readings, stated carefully. First, this is a discipline result: the dramatic regime exists in the model, and the data declines to put the economy in it. Second, the result is portable in a way a numerical scan would not be. The bound $\bar{\sigma}$ needs no structural estimation, only participation rates and a private share, so any researcher with a participation gradient can compute it and ask

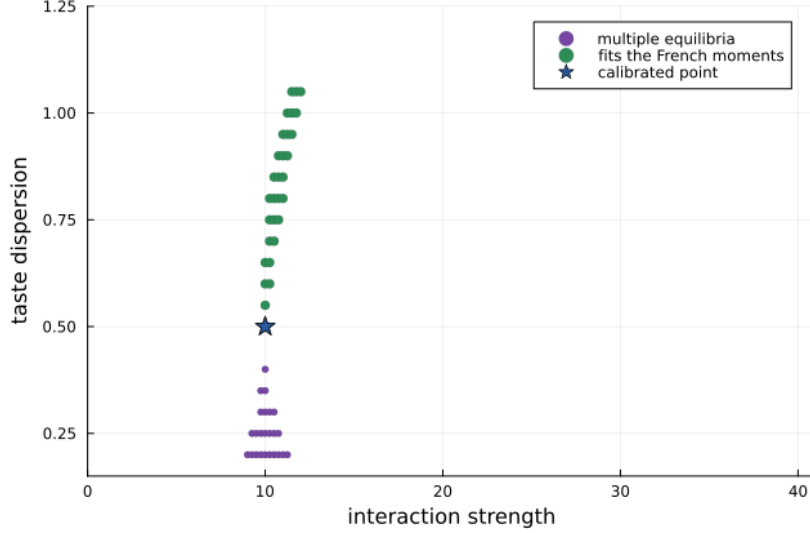


Figure 3: The (κ, σ_m) plane at $\omega = 0.30$. Multiplicity (purple) requires concentrated belonging tastes; fitting the French participation moments (green) requires dispersed ones. The calibrated point (star) sits at the boundary.

whether a coordination-trap story survives their own data. Any account of social tipping that rests on discrete-choice complementarity has to confront the same arithmetic: observed gradients are evidence of taste heterogeneity, and taste heterogeneity is exactly what flattens the aggregate map. Third, the boundary matters as much as the verdict. The calibrated economy sits close to the region, so forces that compress the taste distribution, segregation that homogenises social environments for instance, would push it inside, and the proposition says precisely how much compression it would take. That proximity, not membership, is the model’s honest statement about coordination risk, and it makes the amplification result of the next section the policy-relevant face of the complementarity.

6 Three policies and the social fabric

I run three policies on the calibrated economy, chosen so that the comparison sits where this model’s framework adds the most value. The first is exactly the experiment of the companion paper, rerun on the participation margin so that the two papers can be read side by side: a budget-balanced 20 percent labour-income subsidy. The second is the structural counterpart to that fiscal experiment, an empowerment shock that raises the lower-education group’s agency without a tax. The third is a deliberately recognisable policy that this class of model has not, to my knowledge, evaluated structurally before: a participation tax credit, the analogue of France’s 66 percent charitable-donations deduction, the United Kingdom’s Gift Aid, and the United States itemised deduction. The three policies are designed to sit on opposite sides of the social complementarity, so the same calibrated mechanism produces effects of opposite sign for fiscally similar interventions.

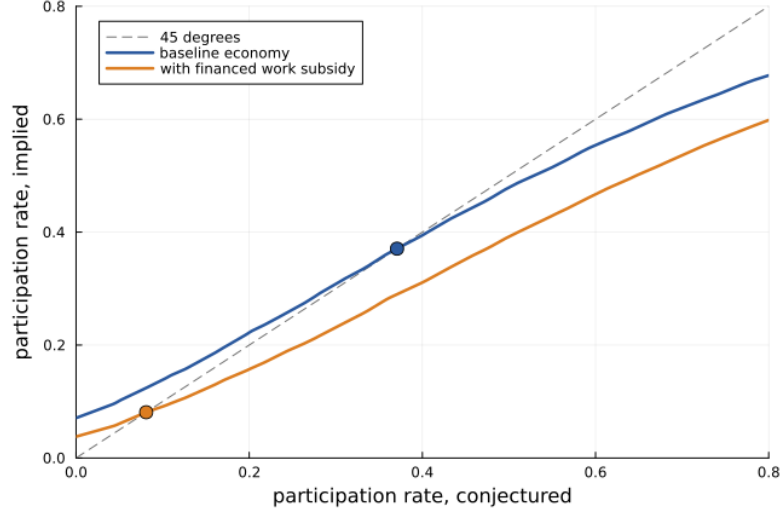


Figure 4: The aggregate participation map at the calibrated point (blue) and under the financed make-work-pay subsidy (orange), with stable equilibria marked. The policy shifts the whole response down; the equilibrium slides from 37 to 8 percent.

6.1 Make-work-pay: an eight-point policy and a twenty-nine-point collapse

The companion paper’s activation experiment, a 20 percent labour-income subsidy financed by a lump-sum tax, runs here on the participation margin. Both arms of the policy push against participation: the subsidy raises the return to market hours, and the tax makes households poorer, which raises the utility cost of the time lump. The budget balances at a lump-sum tax of 0.093, about a fifth of mean labour income.

Figure 4 shows the aggregate map before and after, and Figure 5 the decomposition. The direct effect of the policy, evaluating every household’s participation choice at the baseline belonging level, is a fall from 37.1 to 29.0 percent: eight points, the policy’s true price-and-income effect. The equilibrium effect is a fall to 8.1 percent: twenty-nine points. The ratio is 3.6. The difference is the social feedback: each exit lowers the participation rate, which lowers the belonging return of everyone still in, which causes further exits, the complementarity working as an amplifier rather than as a source of multiplicity. The economy does not jump between equilibria, there is only one; it slides down a steep stretch of a unique fixed point, which is what living near the boundary of the coordination region means in practice.

Honesty about the financing: the lump-sum tax does real work in this result, both because it is large and because it is regressive in utility terms. That is not a bug of the experiment; it is the experiment, the same financed policy as the companion paper, and the decomposition separates what the policy does directly from what the social structure does with it. The amplification factor, not the level, is the portable number.

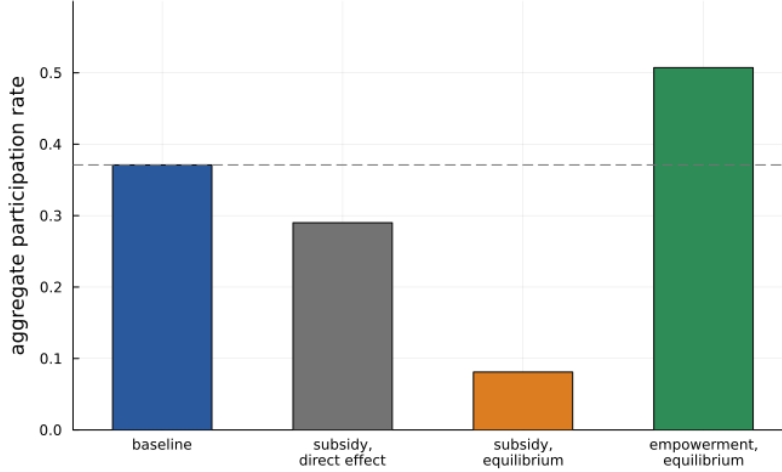


Figure 5: Aggregate participation: baseline, the subsidy’s direct effect, the subsidy in equilibrium, and the empowerment policy in equilibrium. The social feedback multiplies the subsidy’s direct effect by 3.6.

6.2 Empowerment: the same goal, the opposite social consequence

The second policy raises the lower-education group’s agency from 0.765 to the population mean of 0.838, holding everything else fixed: an investment in the capacity to convert effort into income, training, health, labour-market access, with no offsetting tax in this partial-equilibrium experiment. Equilibrium participation rises from 37.1 to 50.7 percent, with the lower-education group at 42 percent, nearly closing its gap, and the higher group at 60 (Figure 5, last bar). The channel is the gradient decomposition’s wealth effect at work: empowered households are richer, the time lump costs them less in utility, they join, and the complementarity multiplies the inflow exactly as it multiplied the outflow before.

6.3 A participation tax credit: the social-feedback amplifier in reverse

The third policy is a recognisable real-world instrument that this model class has not, to my knowledge, evaluated structurally before: a participation tax credit, the model analogue of France’s 66 percent charitable-donations deduction, the United Kingdom’s Gift Aid (a 25 percent basic-rate top-up), and the United States itemised charitable deduction. The credit rebates a fraction ρ of the opportunity cost of the participation time lump, the foregone earnings $\alpha z \bar{q}$, paid directly into the household budget when the household participates and financed by the same lump-sum tax channel as the work subsidy. The rebate rate ρ is the actual policy parameter, not a free dial: $\rho = 0.66$ for the French deduction, $\rho = 0.25$ for Gift Aid. Unlike the work subsidy, which raises the return to hours and so the price of participation time, fighting the complementarity, the credit lowers the net cost of joining and works with it.

Take the Gift Aid rate as the central case, since it is the smaller and better-identified of the two. A 25 percent rebate raises equilibrium participation from 37.1 to 83.3 percent at a fiscal cost

Table 1: Participation credit under imperfect take-up by the lower-education group. The ratio of high to low participation falls from 1.79 at baseline in every scenario: the credit is equalising even when the group that claims least is the one that starts lowest.

	participation	low	high	cost (% of mean income)
baseline	0.371	0.265	0.476	—
Gift Aid 25%, full take-up	0.833	0.760	0.907	4.1
Gift Aid 25%, $\tau = 0.5$	0.823	0.741	0.906	3.2
Gift Aid 25%, $\tau = 0.25$	0.732	0.584	0.880	2.6
France 66%, full take-up	0.997	0.993	1.000	12.7
France 66%, $\tau = 0.5$	0.941	0.881	1.000	9.5
France 66%, $\tau = 0.25$	0.883	0.765	1.000	8.1

of 4.1 percent of mean labour income, roughly a fifth of what the work subsidy costs, with the sign reversed and with both groups strictly interior at 76 and 91 percent. The French 66 percent deduction, a much larger rebate, drives participation to 99.7 percent at a cost of 12.7 percent of mean income. That figure should be read as an upper bound rather than a forecast: at a rebate that large the model saturates, the higher-education group goes to a corner, and the exercise stops being informative about levels while remaining informative about direction. The mechanism in both cases is the social feedback running the desirable way: each induced entry raises the participation rate, which raises every other household’s belonging return, which induces further entries. The same complementarity that amplified the work subsidy’s eight-point direct effect into a twenty-nine-point collapse turns a modest rebate into a large inflow. Implementing the rebate inside the budget constraint, rather than as a multiplier on the social payoff, if anything strengthens it, because the rebate also relaxes the participant’s consumption and the income effect reinforces the social one.

Take-up incidence

Credits of this kind are not claimed evenly. Itemisation, salience and the paperwork of claiming all favour better-off households, so the headline numbers assume a take-up rate the instrument would not achieve. I model this with a take-up ratio τ : the lower-education group’s effective rebate is $\tau\rho$ while the higher group receives the full ρ , and the fiscal cost counts only credits actually claimed. Table 1 runs $\tau \in \{1, 0.5, 0.25\}$.

Two things come out of the exercise, and the first was not what I expected. Imperfect take-up does not make the credit regressive. The ratio of higher- to lower-group participation is 1.79 at baseline and falls in every scenario, to 1.01 when take-up is complete and still to 1.31 and 1.51 when the lower group claims at only a quarter of the higher group’s rate. The credit equalises participation even when the households who claim it least are the ones who start furthest behind, because the lower group has the most room to move and the complementarity carries part of the adjustment for it. Poor take-up makes the instrument less equalising, not regressive.

Second, take-up disciplines the level without touching the sign. Gift Aid’s aggregate falls from 83 to 73 percent as take-up drops to a quarter, and its cost falls from 4.1 to 2.6 percent of mean

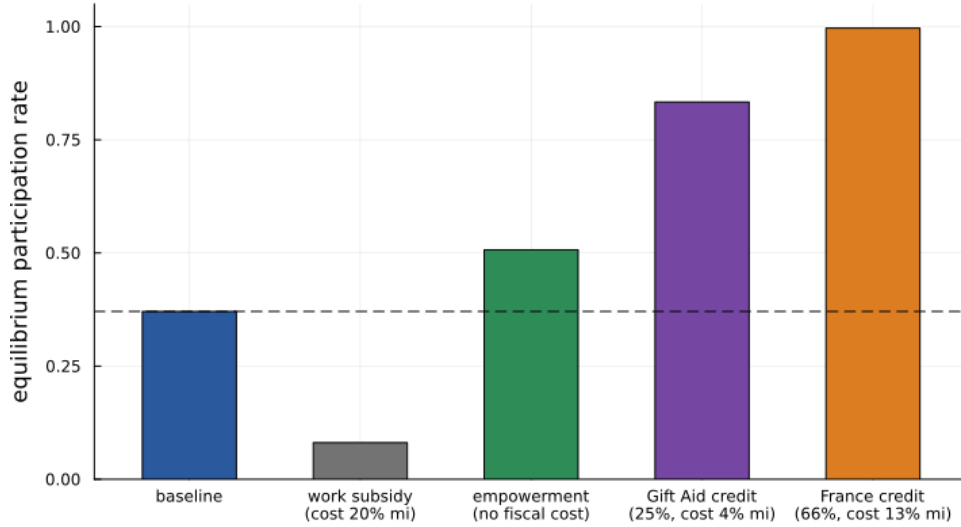


Figure 6: Equilibrium participation under the five scenarios. The work subsidy costs about a fifth of mean labour income and collapses participation to 8 percent. Gift Aid’s 25 percent rebate, the central credit case, costs a twenty-fifth and lifts participation by 46 points with both groups interior; the France-style 66 percent rate costs about an eighth and saturates the model, so it is shown as an upper bound. Empowerment carries no fiscal cost. The credit and the work subsidy sit on opposite sides of the same social complementarity.

income, so the policy stays cheap relative to the work subsidy throughout. The French rate is the exception that proves the earlier caution: because take-up throttles only the lower group while the higher group still receives the full rebate, the higher group sits at a corner whatever τ is, which is why the 66 percent case belongs in the paper as a bound and not as an estimate.

Figure 6 compares all three policies at one glance.

Honesty about the magnitudes. The model’s participation-credit response is large because the credit acts on exactly the margin where the complementarity operates, and the complementarity is what the calibration has just pinned down. The take-up exercise captures one real-world friction and tempers the level without touching the sign; others it does not capture, notably eligibility rules and the gap between the formal contributions a credit scores and the broader participation measured here, all push the same way. The credit numbers are therefore upper bounds at a given rebate intensity, with the Gift Aid case the one to quote and the French rate the ceiling. The portable result is the asymmetry. At a lower fiscal cost than the work subsidy the credit moves participation in the opposite direction and by more; and the social complementarity is the policy lever, not the friction the lever has to fight against, when the policy is designed to work with the complementarity rather than against it.

The three experiments together form the paper’s policy message. All three are pro-work or pro-social policies, but they sit on different margins. A consumption-index evaluation would score the work subsidy positively (consumption rises by five and a half percent) and could not even see

Table 2: Two ledgers for the same policies. Material GDP is mean labour income; GDP-B adds the social fabric priced at the model’s marginal rate of substitution. The two fiscal policies are near mirror images across the ledgers.

	participation	GDP	GDP-B
baseline	0.371	—	—
work subsidy (20%)	0.081	+7.2%	−2.2%
empowerment	0.507	+2.2%	+5.9%
participation credit (66%)	0.997	−6.7%	+12.5%

the social cost; it would miss the empowerment policy’s wellbeing channel; and it would score the participation credit as a pure transfer with deadweight loss, missing the boom entirely. If the social dimension of wellbeing carries weight, as the SAGE framework and the wellbeing literature hold [Lima de Miranda and Snower, 2020, Benjamin et al., 2023, Layard and De Neve, 2023], the three policies are not on a common index; they sit on a plane that the single-index evaluation cannot reach.

6.4 GDP and GDP-B: the same policies on two ledgers

The claim that a single-index evaluation cannot rank these policies can be made precise in the language of the beyond-GDP accounting programme [Brynjolfsson et al., 2019, Stiglitz et al., 2009]. Define material GDP as mean labour income and GDP-B as material GDP plus the value of the social fabric, the participation rate priced at the marginal rate of substitution between cohesion and consumption in experienced wellbeing, evaluated at the baseline. The model’s own price puts a fully participating society’s cohesion at 33 percent of GDP, a number the WELLBY money valuation of wellbeing [Layard and De Neve, 2023] can discipline directly; the qualitative ledger of Table 2 holds for any price above 25 percent of GDP, the point at which the work subsidy’s gain in material output is exactly offset by its loss of fabric.

The two ledgers tell opposite stories about the same fiscal choice. The work subsidy raises material GDP by 7.2 percent and lowers GDP-B by 2.2 percent: it manufactures measured output by dismantling the fabric. The participation credit does the reverse, lowering material GDP by 6.7 percent, because participation takes time that would otherwise be worked, while raising GDP-B by 12.5 percent. Empowerment raises both. The two fiscal policies are near mirror images across the two ledgers (Figure 7), which is the beyond-GDP programme’s point stated inside a structural model: the choice between subsidising hours and rewarding participation is not a matter of degree on one axis but of direction on two.

7 Conclusion

The paper’s central result is a bound. In a discrete-choice model of social participation with lognormal tastes, multiple equilibria require taste dispersion below a threshold built from observed participation rates and one preference share, and that threshold tightens as the participation gradient

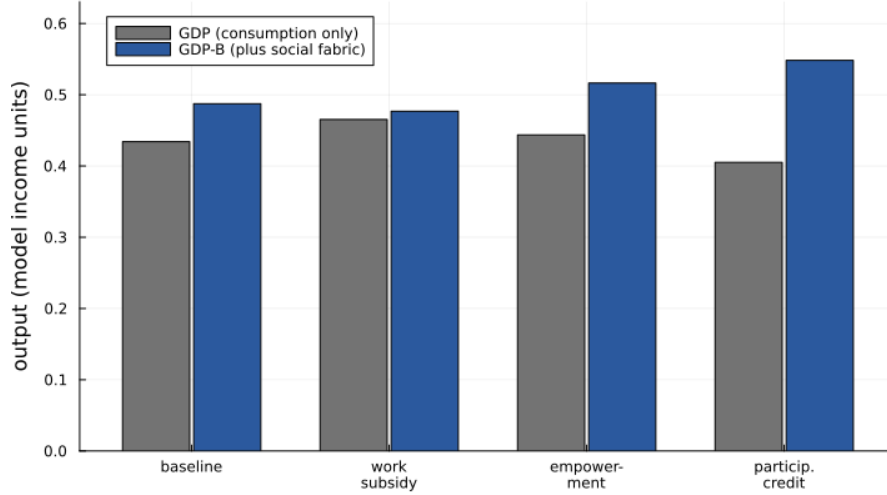


Figure 7: Material GDP (consumption only) against GDP-B (consumption plus the social fabric) for the baseline and the three policies. The work subsidy is the only policy that raises GDP while lowering GDP-B; the participation credit is its mirror image.

steepens. Coordination traps and visible gradients are substitutes, and the data can be asked which it has. Asked of France, it answers gradients: the economy sits about ten percent outside the coordination region, conservatively, since the bound overstates the true slope. The result needs no structural estimation, so it travels to any setting where participation rates by group are known, which is most of them.

Around that bound the paper gives the SAGE agency dimension a structural home on the social side of the model. Agency sets who can afford to participate, through wealth rather than through the price of time, which is why participation rises with income in the model as in the data, and which supplies the dispersed tastes the bound then rules on. Policy lives at the edge the bound locates. The social complementarity amplifies a financed work subsidy’s eight-point direct effect into a twenty-nine-point collapse of participation; it amplifies an empowerment policy into a participation boom of fourteen points without a fiscal instrument; and it amplifies a participation tax credit into a still larger boom at a lower fiscal cost than the work subsidy, with the sign reversed. The same calibrated complementarity does all three pieces of arithmetic. A recognisable real-world policy that pays direct attention to the social margin therefore outperforms the work-side policy by orders of magnitude on its own terms, and the contrast is invisible to single-index evaluation.

The designed extensions are unchanged and now better motivated: a general-equilibrium close with transitions [Auclert et al., 2021], so that the collapse and the boom have price feedbacks and dynamics; the estimation of the cohesion weight from life-satisfaction panels, which would let the participation results be priced in wellbeing units; and the environmental dimension, completing the framework. The engine is shared, the data targets are public, and every result in both papers is reproducible from the repository scripts.

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